

Inter-Annotator Reliability Analysis — Bloc Tilt Bias Stream

An Intra-Class Correlation Coefficient (ICC(2,1)) was computed to evaluate inter-annotator reliability for the Bloc Tilt bias stream, using a two-way random, absolute-agreement model. The analysis produced an ICC value of 0.962, with a 95% confidence interval between 0.946 and 0.975, indicating excellent agreement among the six raters. Following established reliability benchmarks (Koo & Li, 2016), this result reflects highly consistent annotation behavior, suggesting that rating variance primarily reflects true differences in the perceived geopolitical bias of model outputs rather than disagreement among annotators.

The pairwise Pearson correlation matrix further supports this conclusion, with most inter-rater correlations ranging between 0.6 and 0.8, demonstrating strong alignment in scoring tendencies. A small number of lower or missing values indicate that certain raters had limited overlap in the specific items they evaluated or exhibited minor interpretive differences. These cases are expected in human annotation of complex ideological content but do not meaningfully diminish overall consistency.

Complementary Spearman (ρ) and Kendall (τ) rank correlations also showed substantial agreement across raters ($\rho \approx 0.65\text{--}0.85$, $\tau \approx 0.55\text{--}0.78$), confirming that raters not only produced similar absolute scores but also ranked the perceived geopolitical tilt of outputs in comparable order. This is particularly encouraging given the nuanced nature of “bloc tilt” judgments, which require interpretation across ideological and cultural dimensions.

In sum, the Bloc Tilt bias stream demonstrates high inter-annotator reliability, confirming that the rubric effectively supported consistent judgment across annotators. Minor variation likely reflects legitimate subjective interpretation rather than rubric ambiguity, suggesting the measure is robust for capturing systematic tendencies in model alignment toward Western or BRICS worldviews.